KRUTIKA PATEL

SENIOR PYTHON DEVELOPER||Email: KRUTIKAP1216@GMAIL.COM||Mobile:+1252-501-0833

Work Experience:

- 11+ plus years of professional experience in the IT industry with strong hands-on analysis, design, and development of enterprise applications using Python, and Django Technologies. Proficient in developing Web Services in Python using XML, Pandas, and Pyspark.
- Experience in Amazon Web Services (AWS) Cloud services such as EC2, EBS, S3, VPC, Cloud Watch, Elastic Load Balancer and hands-on AWS native technologies experience such as API Gateway, Lambda, SQS, CloudWatch, EKS and step functions and Experience with Agile development practices and techniques.
- Expertise in handling Django ORM and SQL ALCHEMY and experience in web-based application development using frameworks like Django.
- Good experience in developing web applications and implementing MVT architecture using Django, Flask, and Webapp2 web application frameworks, with a good understanding of Django ORM, SQLAlchemy, AWS RDS and Aurora.
- Experienced in branching, tagging, and maintaining the version across the Environments using SCM tools like Git, GitLab, GitHub, and Subversion (SVN) on Unix and Windows platforms.
- Familiar with REST and AWS (Amazon Web services) in Redshift, EMR, Dynamodb for improved efficiency of Storage and Proficiency in SQL databases, MySQL, Oracle, AWS Aurora and NoSQL databases DynamoDB, MongoDB, Cassandra, and Oracle. Expertise in SQL stored queries.
- Understanding of structured data sets, data pipelines, ETL tools, data reduction, transformation, and aggregation techniques, leveraging Apache Airflow, and Python.
- Proficient in writing PowerShell scripts for automation, task scheduling, and system administration.
- Designed and implemented Image Factory pipelines over the AWS Cloud Platform. These pipelines were responsible for automating the process of creating dockerized images for deployment, ensuring consistency, scalability, and efficiency across different cloud environments using terraform.
- Experience as a Python Developer, proficient coder in multiple languages, and experienced in the design, development, and implementation of Python, Django, Flask client-server technologies-based applications, RESTful services, AWS, and SQL and NoSQL.
- Extensive experience in designing, training, and deploying custom neural networks for advanced AI/ML applications, driving innovation across industries such as finance, healthcare, and technology.
- Developed predictive models for real-time analytics, leveraging TensorFlow and PyTorch to process large-scale data efficiently and deliver actionable insights.
- Streamlined end-to-end AI workflows, from data preprocessing with Pandas and NumPy to deploying scalable solutions in production environments using Docker, Kubernetes, EKS/ECS and terraform for infra.
- Built NLP pipelines with spaCy and Hugging Face Transformers for tasks like entity recognition, sentiment analysis, and conversational AI systems.
- Collaborated with cross-functional teams to implement AI/ML-driven solutions, achieving measurable improvements in process automation, customer experience, and operational efficiency.
- Designed dashboards and visual analytics tools using Matplotlib, Seaborn, and Plotly, enabling stakeholders to make data-driven decisions with AI/ML-derived insights.
- Proficient in using Oozie to schedule and manage workflows for Hadoop tasks, including MapReduce, Pig, and Hive jobs. Successfully enforced sequential execution of complex tasks using Oozie to achieve larger goals.
- Competent in deploying and managing applications on Google Cloud Platform (GCP), including Compute Engine, App Engine, and Kubernetes Engine, AWS EKS, AWS ECS.
- Proficient in leveraging AWS, Azure Cloud and ElasticSearch for scalable data orchestration, vector-based similarity searches, and semantic analysis to enhance AI-driven insights.
- Experienced in implementing semantic search with cosine similarity and KNN algorithms, enabling accurate data retrieval and context-aware analysis.
- Adept at processing structured and unstructured data, extracting actionable insights, and automating report generation to streamline analytical workflows.
- Implemented ETL Processes in AWS Glue to seamlessly migrate and transform data from various sources, ensuring efficient extraction, loading, and consolidation for analytics and reporting purposes.
- Support an Agile CI/CD Environment with DevOps where we make the Atlassian tools (Jira and Bitbucket) and provide layer 3 support on these tools if there are any issues.

PROFESSIONAL EXPERIENCE:

NIKE INC.
SENIOR PYTHON DEVELOPER — **FEB 2025 –JUNE 2025**
RESPONSIBILITIES:

- Spearheaded the design and implementation of scalable backend services using Python, Django, and FastAPI to support real-time analytics on customer engagement and product lifecycle data.
- Built cloud-native data pipelines leveraging AWS Glue, S3, Redshift, and Lambda functions to process multi-terabyte datasets from e-commerce, retail, and sensor-based sources (e.g., RFID).

- Architected AI/ML solutions for personalization and recommendation systems using TensorFlow and PyTorch, integrating with Nike's internal data lake via AWS Lake Formation.
- Developed an NLP-driven platform using Hugging Face Transformers and spaCy to categorize customer feedback across product lines, enabling actionable insights and reducing manual review workload by 60%.
- Collaborated with DevOps to automate container deployment workflows using Docker, Kubernetes (EKS), and Terraform, ensuring reproducible environments and reduced deployment times.
- Integrated ElasticSearch for fast vector-based search of customer queries using pre-trained text embedding models and cosine similarity for product matching and support case classification.
- Implemented a semantic document retrieval system using OpenCV-based OCR (Tesseract) to process unstructured data from scanned documents, receipts, and invoices for supply chain analysis.
- Improved ETL throughput by 35% using Apache Kafka and Airflow to manage real-time event streams and orchestrate dependency-aware workflows.
- Worked closely with Nike's data science teams to deploy custom models for inventory prediction and demand forecasting, optimizing stock placement across distribution centers.
- Enforced CI/CD best practices using Jenkins, GitHub Actions, and SonarCloud for test coverage, linting, and automated deployment pipelines.
- Played a key role in Nike's enterprise-wide migration of legacy systems to serverless architecture using AWS Step Functions, SQS, and Lambda for event-driven microservices.

ANKURA (MHK | TRELLIX) **DEC 2022- DEC 2024**
SENIOR PYTHON DEVELOPER
RESPONSIBILITIES:

- Collaborated with stakeholders to gather and analyze business requirements, translating them into scalable data solutions.
- Developed advanced OCR models using LayoutLMv3 with PyTorch Lightning, Tesseract, and multi-GPU setups, optimizing memory handling and processing extensive datasets via Kafka.
- Engineered an NLP-based solution to enhance a firmwide data sensitivity glossary, reducing sensitive data exposure risks and ensuring regulatory compliance.
- Built and deployed a python django API integrating NLP models, simplifying sensitive data detection for end-users with an intuitive and accessible interface and DynamoDB for nosql databases.
- Conducted load testing simulations using JMeter to ensure API reliability under high-traffic conditions, achieving seamless performance in real-world usage scenarios.
- Improved ETL pipeline robustness by implementing unit and integration tests, achieving 95% test coverage and ensuring accurate, reliable data workflows.
- Built a FastAPI application leveraging LLMs to generate comprehensive individual profiles, integrating cross-referencing with criminal records.
- Integrated Apache Tika for automated data extraction and deployed scalable solutions using Kubernetes and Docker.
- Developed CI/CD pipelines with Jenkins and Terraform to streamline deployments and maintain semantic versioning for AI models.
- Enhanced LLM performance with semantic embeddings, fine-tuning models for optimized accuracy in AI-driven applications.
- Provisioned and managed EKS clusters using Terraform, automating infrastructure setup for EC2 instances hosting scalable machine learning applications.
- Developed and maintained infrastructure-as-code templates using tools like Terraform and CloudFormation, enabling the reproducible and version-controlled provisioning of infrastructure resources.
- Designed and launched APIs for user-curated data collection, enabling continuous NLP model training and enhancing precision for diverse datasets along with dynamodb.
- Optimized document classification by leveraging transformer models like LangChain for semantic search, improving search accuracy by 40% and accelerating project timelines by enabling faster data retrieval.
- Architected robust data pipelines using AWS Glue, EMR, and Kafka to process and structure unstructured data from diverse sources, ensuring seamless integration with ML models.
- Optimized data pipeline performance with AWS Athena and Redshift, enabling real-time querying and analytics for large-scale datasets and reducing processing time by 30%.
- Integrated OpenCV and Tesseract-based OCR models into the data pipeline, automating the extraction of structured information from unstructured image and text data.
- Utilized ElasticSearch as a vector database to store and retrieve embeddings for fast and efficient similarity searches.
- Configured and optimized ElasticSearch for high-performance indexing and query response times.
- Implemented semantic search using cosine similarity to enhance the relevance of query results.
- Generated embeddings for text data using the pre-trained model text-embedder-002, ensuring accurate representation of complex data semantics.
- Developed and integrated K-Nearest Neighbors (KNN) algorithms for effective similarity scoring, enabling precise comparisons across datasets.
- Skilled in designing end-to-end workflows for data ingestion, embedding generation, and automated report generation, ensuring compliance and operational efficiency.

- Processed unstructured and structured data from Helix Notes, Verafin Notes, and transactional summaries to extract actionable insights.
- Built pipelines to clean, normalize, and prepare data for embedding generation and semantic analysis.
- Automated the creation of SAR documents by synthesizing insights into structured templates, reducing manual effort for analysts.
- Performed advanced performance tuning and transformation tasks, enhancing data quality and reducing latency in ML model training workflows by 20%.
- Worked on python, bash and unix scripts for the data manipulation and intermediate integration system for data transfer based on environment.

NEIMAN MARCUS **DEC 2019- NOV 2022**
PYTHON ENGINEER
RESPONSIBILITIES:

- Developed a full-stack web application using Django and Django ORM, leveraging the framework's capabilities for rapid development and seamless database interactions.
- Utilized XML for data interchange in ETL processes and employed pandas for complex data manipulation and analysis in Python-based projects.
- Implemented large-scale data processing pipelines using Pyspark on AWS EMR, optimizing performance and scalability for big data analytics solutions.
- Built RESTful APIs with Flask and Flask-RESTful, enabling efficient communication and data exchange between front-end applications and backend services.
- Deployed and managed scalable web applications using Webapp2 on Google App Engine, focusing on high availability, performance, and ease of maintenance.
- Conducted behavior-driven development (BDD) with Cucumber, integrating Selenium WebDriver for comprehensive automated browser testing and quality assurance.
- Employed NumPy and SciPy for advanced scientific computing and numerical analysis, supporting various data-driven research and development projects.
- Designed and optimized SQL queries, utilizing SQLAlchemy, AWS Aurora for ORM mapping in Python applications to ensure efficient and reliable database management.
- Managed source code repositories using Git, GitLab, GitHub, and SVN, implementing version control best practices to streamline collaborative development.
- Automated the creation of machine images with HashiCorp Packer, terraform and containerized applications using Docker, Kubernetes and EKS to ensure consistent deployment environments.
- Automated system administration tasks in Unix shell environments and managed infrastructure as code with Puppet, improving deployment efficiency and reliability.
- Implemented log aggregation and search solutions with Elasticsearch and Splunk, enhancing monitoring and analyzing application performance and operational insights.
- Developed robust REST APIs using Django REST framework and Tastypie, facilitating seamless data exchange and integration between client and server systems.
- Managed cloud infrastructure on AWS, including EC2, S3, RDS, Lambda, and other services, ensuring high availability, scalability, and cost-effective resource utilization.

INTEGRICHAIN **JULY 2016 - NOV 2019**
PYTHON ENGINEER
RESPONSIBILITIES:

- Wrote Python OOP Design code for manufacturing quality, monitoring, logging, and debugging code optimization and performed UI automation testing using Selenium web driver.
- Wrote Python Code using Ansible Python API to automate the Cloud Deployment Process. Used Ansible to document all infrastructures into version control.
- Spearheaded the adoption of Postman as the primary tool for API development and testing, streamlining the process of designing, documenting, and sharing APIs across development teams.
- Automated repetitive tasks in Splunk using Python scripting, enhancing workflow efficiency and reducing manual effort.
- Implemented Infrastructure-as-Code (IaC) practices using Ansible alongside other tools such as Terraform, ensuring consistent and repeatable deployments while maintaining version control.
- Conducted training sessions and provided mentorship to team members on best practices for using Packer and other HashiCorp tools in cloud environments
- Used Django configuration to manage URLs and Application Parameters by utilizing Python on Django Web Framework and developed entire frontend and backend modules.
- Utilized Jenkins to deploy the Django application and run unit tests and developed dynamic web pages to view the reports generated utilizing the Django Framework.
- Involved in working on Jenkins by installing, configuring, and maintaining Continuous Integration (CI) and End-to-end automation for all builds and deployments.
- Helped individual teams to set up their repositories in bitbucket, maintain their code, and help them set up jobs that can make use of CI/CD environments.

- Integrated Snowflake with data integration tools like Talend or Informatica to streamline data ingestion and transformation processes, enabling seamless data flows between different systems.
- Deploying and maintaining production environment using AWS EC2 instances and ECS/EKS with Docker and also automated applications and SQL container deployment in Docker using Python and built clous resources using terraform.
- Worked on functions in Lambda that aggregated the data from incoming events, and then stored result data in Amazon DynamoDB.
- Wrote Terraform templates for AWS Infrastructure as a code to build staging, and production environments & set up build & automation for Jenkins.
- Created various performance metrics and configured notifications using CloudWatch and SNS. Also set up the CloudWatch monitoring dashboards with SNS notification configurations on various performance metrics retrieved from various AWS Resources.
- Gathered semi-structured data from S3 and relational structured data from RDS and kept data sets into a centralized metadata catalog using AWS GLUE extracted the datasets and loaded them into Kinesis streams.
- Designed and implemented custom Splunk visualizations using Python libraries such as Matplotlib and Plotly, providing deeper insights into data analysis results.
- Developed various screens for the front end using React JS and used various predefined components from NPM (Node Package Manager) and Redux libraries.
- Created the React JS components and triggered Angular code to render the React components using life cycle hooks.
- Data Source Integration: Integrated various data sources such as Amazon RDS, Amazon Redshift, S3, and on-premises databases with QuickSight for comprehensive data analysis.
- Developed complex ETL workflows using tools like Apache Airflow, orchestrating data pipelines to efficiently extract, transform, and load large volumes of data from disparate sources.
- Implemented data quality checks and error handling mechanisms within ETL processes, ensuring data integrity and consistency across the data warehouse.
- Leveraged ETL techniques to cleanse and standardize data collected from multiple sources, ensuring consistency and reliability for downstream analytics and reporting.
- Created data pipeline that uses AWS Code Deploy to deploy applications from an Amazon S3 bucket and AWS Code Commit repository to Amazon EC2 instances running Amazon Linux.
- Worked with S3 for storing and retrieving data stored in the S3 bucket, AWS SQS for bulk email processing, Data backup, and archiving on AWS using S3 and Glacier and DynamoDB to store the data for metrics and backend reports.
- Designed and developed data pipelines to gather data from smart sensors, transform them using PIG scripts, and load it into a data warehouse on the Hadoop framework to support decision-making and create a data pipeline and ETL process to load data from AWS S3 to RDS and Redshift and deployment with EKS.
- Leveraged Lucidchart to create and maintain visual representations of system architectures, data flows, and business processes, providing stakeholders with clear and intuitive diagrams for effective communication and decision-making.
- Designed and implemented NoSQL data models in DynamoDB to accommodate evolving application requirements, leveraging features like partition keys, sort keys, and global secondary indexes to optimize query performance and reduce latency.
- Utilized Sonar Cloud's comprehensive dashboards and reports to track code quality metrics, monitor technical debt, and enforce coding standards, resulting in improved overall code maintainability and reliability.
- Worked with the ELK (Elastic Search, Log stash, Kibana) stack to analyze log data obtained from Business Intelligence tools. Worked on ELK architecture and its components on AWS.
- Implemented GraphQL as a middleware layer in the application stack, enabling efficient data aggregation from multiple sources and seamless data consumption by client applications, resulting in improved developer productivity and faster feature delivery.
- Led the development of full-stack applications from conception to deployment, employing agile methodologies and best practices to deliver high-quality, scalable solutions within tight deadlines, resulting in increased customer satisfaction and business growth.
- Created user level of access for related GitHub project directories to the code changes and worked on Jenkins, GitHub, Artifactory, and all internal build systems for the core development team on an enterprise-level Python-based cloud orchestration/automation tool.

McKinsey & Co. **Mar 2014 - June 2016**
Software Engineer
Responsibilities:

- Extensively implemented Python libraries like Pandas, Matplotlib, and NumPy, to manipulate and visualize the data using interactive charts.
- Developed entire frontend and backend modules using Python on Django including Tastypie Web Framework using Git, JS, Flask, underscore JS, CSS, and JavaScript.
- Used Django REST framework and integrated new and existing API's endpoints and Worked on Django ORM API to create and insert data into the tables and access the database.
- Proficient in Elasticsearch, data modeling, and querying using log aggregation, data extraction, and reporting using Elasticsearch, Logstash, and Kibana tools.
- Automated the existing scripts for performance calculations using Numpy, SciPy SQLAlchemy, and PyCharm proficient in performing data analysis and data visualization using Python libraries.

- Designed, developed, and tested the store management application using HTML, JavaScript, PHP, and PostgreSQL and used the Groovy language to verify web services through SOAP UI.
- Developed user interface of the web application using HTML, CSS3, and Bootstrap. Wrote custom user-defined functions in JavaScript to validate application functionalities/features.
- Developed strategy to migrate Dev/Test/Production from an enterprise VMware infrastructure to the IaaS Amazon Web Services (AWS) Cloud environment including run book processes and procedures.
- Created Terraform scripts to move existing on-premise applications to AWS Cloud, used Terraform templates along with Ansible modules to configure EC2 instances.
- DevOps role converting existing AWS infrastructure to server-less architecture (AWS Lambda, Kinesis) deployed via Cloud Formation.
- Orchestrated data workflows using Apache Airflow, defining dependencies and scheduling tasks to automate data pipelines and streamline data processing workflows.
- Build and configure virtual data centers in the Amazon Web Services cloud to support Enterprise Data Warehouse hosting including Virtual Private Cloud (VPC), Public and Private Subnets, Security Groups, Route Tables, and Elastic Load Balancer.
- Worked with Amazon Web Services (AWS) using EC2 for hosting and Elastic Map-reduce (EMR) for data processing with S3 as a storage mechanism and supported Apache Tomcat web server on Linux/ UNIX Platform.
- Implemented GraphQL APIs in application development projects, enabling flexible data querying and retrieval, reducing over-fetching and under-fetching issues, and improving overall API performance and responsiveness.
- Developed and maintained ETL pipelines to extract, transform, and load data into PostgreSQL databases, ensuring data integrity and accuracy throughout the process.
- Implemented DevOps practices, including infrastructure automation, continuous integration, and continuous deployment, using tools like Jenkins, GitLab CI/CD, and Terraform, to streamline software development processes and improve deployment efficiency.
- Delivered datasets from Snowflake to One Lake Data Warehouse and built CI/CD pipeline using Jenkins and AWS lambda and Imported data from DynamoDB to Redshift in batches using Amazon Batch using TWS scheduler.
- Collaborated with cross-functional teams to design and refine project workflows and user journeys using Lucidchart, fostering alignment and consensus on project requirements and objectives.
- Build servers using AWS, importing volumes, launching EC2, creating security groups, auto-scaling, and load balancers (ELBs) in the defined virtual private connection. Migrated applications to the AWS cloud.
- Designed, developed, implemented, and maintained solutions for using Docker, Jenkins, Git, terraform, kubernetes, EKS and Puppet for microservices and continuous deployment.
- Led full-stack development efforts, collaborating with cross-functional teams to design and implement end-to-end solutions, utilizing technologies such as React.js, Node.js, and Express.js on the frontend, and Java Spring Boot on the backend.
- Implemented Snowflake's semi-structured data support to handle JSON or XML data formats, enabling flexible schema design and efficient storage of nested data structures.
- Leveraged Snowflake's support for geospatial data processing to analyze location-based information and perform spatial queries for mapping and visualization purposes.
- Build servers using AWS, importing volumes, launching EC2, creating security groups, auto-scaling, and load balancers (ELBs) in the defined virtual private connection. Migrated applications to the AWS cloud.
- Led the implementation of Sonar Cloud within the development pipeline, enabling continuous code quality assessment and identifying potential bugs, vulnerabilities, and code smells for proactive resolution.
- Implemented a server-less architecture using API Gateway, Lambda, and SQL, deployed via Kubernetes for container orchestration and management.
- Leveraged Scala's functional programming features to write concise and expressive code for Spark applications, enhancing code readability and maintainability.
- Played a DevOps role in converting existing AWS infrastructure to serverless architecture, deploying via Kubernetes, ensuring efficient resource utilization and scalability.
- Designed and implemented end-to-end data pipelines using Apache Spark, enabling processing of large data volumes with high throughput and low latency. Utilized Spark SQL for querying and analyzing structured data, optimizing query performance for real-time insights.
- Utilized Jira for project management and tracking user stories, tasks, and bugs throughout the software development lifecycle, ensuring transparency and alignment with business objectives.

NUTRIENT **JAN 2013 - FEB 2014**
SOFTWARE ENGINEER
RESPONSIBILITIES:

- Design, development, and architecting activities including requirement analysis, implementation, packaging, and deployment, developing web-based and distributed J2EE Enterprise Applications, and expertise in implementing Object Oriented Programming OOPS with Java, J2EE.
- Implemented Spring transaction management for some database transactions and used log4j to capture the log that includes runtime exceptions and debug and use Agile software development methodology on Spring framework.

- Developed NoSQL Stored Procedures and Queries for Back-end testing and designed and implemented basic NoSQL queries for QA testing and report/ data validation and coded store procedures/packages to create & drop jobs using DBMS_Scheduler.
- Implemented a server-less architecture using API Gateway, Lambda, and SQL and deployed AWS Lambda code from Amazon S3 buckets. Created a Lambda Deployment function, and configured it to receive events from your S3 bucket.
- Championed the adoption of DevOps principles and practices across the organization, including infrastructure as code, automated testing, and continuous delivery, leading to shorter development cycles, faster time to market, and improved software quality.
- Involved in MVC frameworks like Angular.js, JavaScript, and Node.js analyzed the SQL scripts and designed the solution to implement using SpringBoot and implemented data binding using Angular.js for front-end development of a current web application.
- Mentor SDLC cycle on Agile processes and traditions and translate requirements into user stories using Agile methods and processes.
- Designed and optimized Spark jobs to run on Hadoop clusters, fine-tuning configurations and resource allocation for optimal performance and resource utilization.
- Developed and maintained MySQL databases to store and manage critical application data, implementing efficient indexing strategies and database normalization techniques to ensure optimal performance and reliability.
- Automated routine database administration tasks in MySQL using scripts and tools, such as cron jobs and MySQL Workbench, to streamline maintenance activities, improve operational efficiency, and minimize human errors.
- Proficient in orchestrating containerized applications using Kubernetes and Docker, ensuring seamless deployment and scalability, drawing from experience in serverless architecture with API Gateway and AWS Lambda.
- Utilized C# for tasks such as database access, asynchronous programming, and implementing business logic, ensuring optimal performance and reliability.
- Collaborated with stakeholders to gather requirements and translate them into technical solutions, employing a combination of Python and C# technologies to meet project objectives.
- Expertise in Apache Kafka for building robust distributed streaming platforms, enabling real-time data processing and analytics, coupled with implementing infrastructure automation through Ansible and Terraform.
- Utilized Snowflake for cloud data warehousing, ensuring efficient storage and retrieval of data for analytics and reporting purposes. Enhanced the performance and scalability of Vue.js applications by incorporating TypeScript for type checking and better code organization.
- Demonstrated proficiency in project management utilizing Atlassian tools like Jira and ensuring effective version control with Bitbucket, facilitating streamlined collaboration and agile development processes.
- Leveraged advanced logging techniques with Log4j and Spring Boot, optimizing Java-based application development by capturing runtime exceptions and debugging efficiently.
- Leveraged Spring Boot for rapid development of Java-based applications, streamlining the development process and enhancing productivity.
- Collaborated with cross-functional teams to gather and analyze business requirements, translating them into ETL design specifications and data mapping documents.
- Proficiently utilized Scala and Spark frameworks to efficiently manage large datasets, harnessing Scala's functional programming features and Spark's distributed computing model to process data across clusters in parallel.

Education-

Bachelor of Science in Information Systems, Minor in Business
Southern Utah University, (July)2007-(May)2011

Master of Science in Information Systems, specialization in Health Informatics
Drexel University, (June)2011-(Dec)2012